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Assignment : 3

Q1. Construct leftmost and rightmost derivation & parse tree for 001100 for grammar:

$S \to 0AS \mid 0$

$A \to S \mid A \mid SS \mid 10$

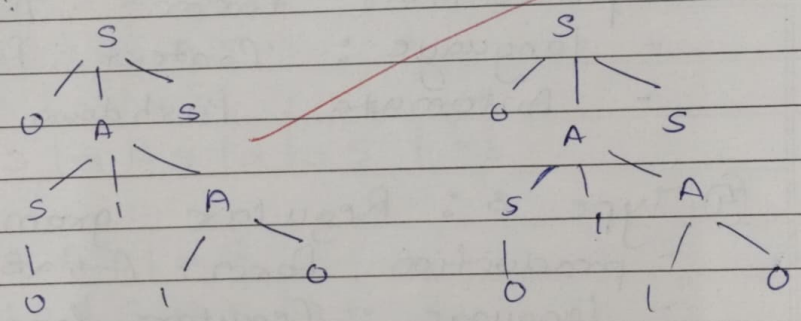
⇒ Leftmost

1. S
0AS
0S1AS
001AS
0011S
001100

Rightmost:

S
0AS
0A0
0S1A0
0S1100
001100

Parse tree :



Q2. Check whether given grammar G is ambiguous

$E \to E+E$

$E \to E-E$

$E \to id.$

Contract string id+ id -id

⇒ $E \to E+E$
id + E-E
id + id - id.

$E \to E-E$
E+E - id
id +id - id.

Grammer is ambiguous.

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Q3. Describe 'Chomskey Hierarchy of language with diagram.

→ ① Type 0: Unrestricted grammer.
- No restriction on production
- Recognized by Turing machine.
- Most powerful.

② Type 1: Context sensitive grammar.
- Production Form: $\alpha AB \rightarrow \alpha \gamma \beta$
- Recognized by linear bounded automata
- language = Context sensitive language.

③ Type 2: Context free grammar.
- production form: $A \rightarrow \gamma$
- language: Context free language
- Automata: Pushdown automata.

④ Type 3: Regular grammar
- production form: $A \rightarrow aB$ or $A \rightarrow a$
- language: Regular language
- Automata: finite state automata.

Q5. Consider given grammar G

$S \rightarrow ASB$
$A \rightarrow aAS \mid a \mid \epsilon$
$B \rightarrow SbS \mid A \mid bb \mid \epsilon$

$\therefore S_0 \rightarrow S$
$S \rightarrow ASB \mid SB \mid AS \mid S$
$A \rightarrow aAS \mid a \mid aS$
$B \rightarrow SbS \mid A \mid bb$

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Now, $B \to A$.

$S_0 \to S$
$S \to ASB \mid SB \mid S \mid AS$
$A \to aAS \mid a \mid aS$
$B \to SbS \mid aAS \mid a \mid aS \mid bb$

$S_0 \to S$
$S_0 \to ASB \mid SB \mid S \mid AS$
$S \to ASB \mid SB \mid S \mid AS$
$A \to aAS \mid a \mid aS$
$B \to SbS \mid aAS \mid a \mid aS \mid bb$.

Remove $S \to S$ & $S_0 \to S$
$S_0 \to ASB \mid SB \mid AS$
$S \to ASB \mid SB \mid AS$
$A \to aAS \mid a \mid aS$
$B \to SbS \mid aAS \mid a \mid aS \mid bb$

$\Rightarrow A \to aAS \mid aS$
$B \to SbS \mid GAS \mid aS$

terminals a and b exists on RHS with non-terminals remove them ~~as~~ from RHS.

$S_0 \to AS \mid ASB \mid SB$
$S \to AS \mid ASB \mid SB$
$A \to XAS \mid XS \mid a$
$B \to SYS \mid YY \mid XAS \mid XS \mid a$.

$S_0 \rightarrow ASB$

$S \rightarrow ASB$, RHS has more than 2 symbols,

Remove it.

$P \rightarrow AS$

$T \rightarrow SY$

$S_0 \rightarrow AS \mid PB \mid SB$

$S \rightarrow AS \mid PB \mid SB$

$A \rightarrow XP \mid YS \mid a$

$B \rightarrow TS \mid YY \mid XP \mid a$.

Gupti
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